LECTURE 1

Foundations of the Internet, Computer Networks & Web Development

Understanding what happens before HTML

COURSE	INSTRUCTOR
Web Development Bootcamp **Cohort 1 (Front-End)**	**Newton Katete**

Why this lecture comes first

Before a learner can understand what HTML, CSS, or JavaScript are doing, it helps to understand the environment in which websites live. A website is not simply “a file on the Internet.” A browser, a network connection, DNS, IP addresses, HTTP, servers, and sometimes databases all cooperate to deliver the page a user sees. This lecture builds that mental model first.

COURSE CONTEXT

Cohort 1 will focus on front-end development: Semantic HTML, CSS, and JavaScript. After the front-end cohort and a two-week break

Learning Outcomes

By the end of this lecture, a learner should be able to:

- Explain what a computer network is and distinguish a LAN, WAN, and the Internet.
- Differentiate the Internet from the World Wide Web.
- Identify the roles of clients, browsers, servers, routers, ISPs, DNS, and IP addresses.
- Describe, at a high level, how TCP/IP and packets move data between devices.
- Explain the purpose of HTTP and HTTPS in browser–server communication.
- Break down the major parts of a URL.
- Describe the sequence of events that occurs when a user opens a website.
- Explain where front-end and back-end web development fit in the client–server model.
- Connect HTML, CSS, JavaScript, PHP, and MySQL to their roles in a web application.

BIG IDEA

Web development sits on top of networking. Networking provides the path; web protocols define how browser and server communicate; web developers create the content and logic that travel over that path.

1. What Is a Computer Network?

A computer network is a group of two or more computing devices connected so they can exchange data and share resources. The devices might be laptops, phones, servers, printers, smart televisions, routers, or many other types of digital equipment.

Examples you already use every day include:

- Your phone and laptop connected to the same Wi-Fi router.
- Computers in a university computer laboratory connected through switches and Wi-Fi access points.
- A mobile phone connected to a cellular provider such as a 4G/5G network.
- A company connecting offices in different towns through wide-area links.

1.1 Why networks exist

Networks make communication and resource sharing possible. A device can request a file from another device, send a message, stream video, access a database, print to a shared printer, or contact a web server located in another country. For web development, the most important idea is that the browser and the web server are usually on different devices and must communicate across a network.

1.2 Common network scopes

Type	Meaning	Typical size	Example
PAN	Personal Area Network	A few metres	Phone connected to earbuds via Bluetooth
LAN	Local Area Network	Room, home, office, campus building	Home or lab Wi-Fi/Ethernet network
WAN	Wide Area Network	Large geographical area	A company network connecting multiple cities
Internet	Global network of interconnected networks	Worldwide	The public Internet

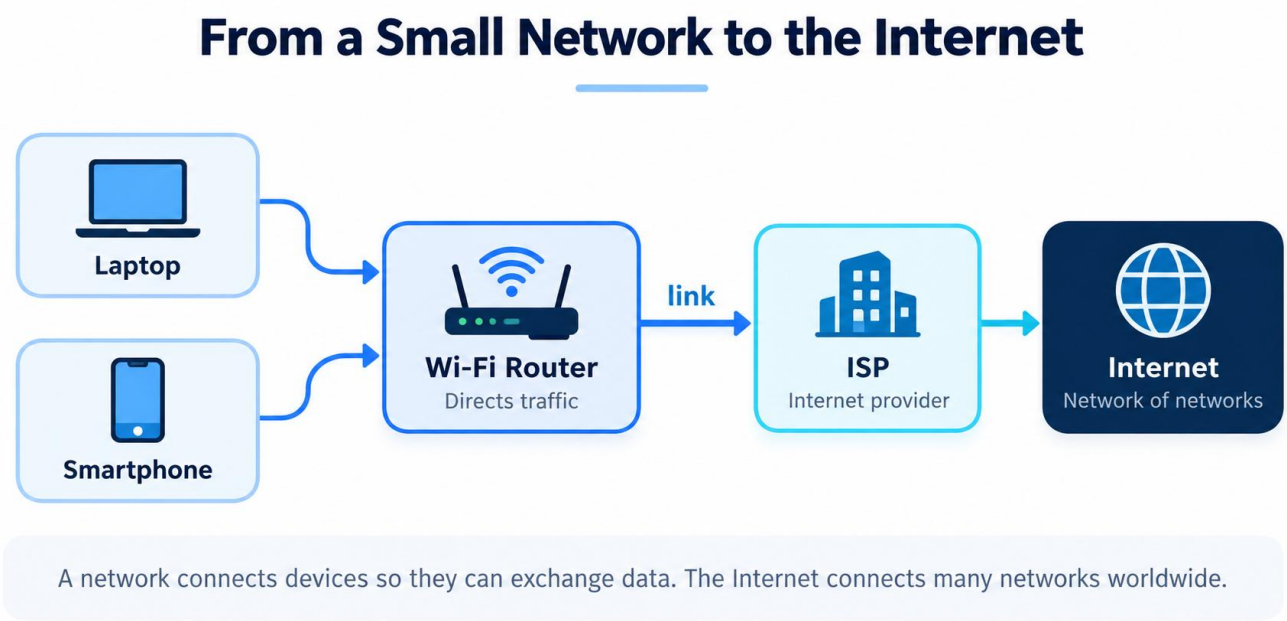


Figure 1. A local device reaches the wider Internet through a router and Internet Service Provider (ISP).

TEACHING ANALOGY

Think of a network like a road system. Devices are buildings, links are roads, routers are intersections that choose the next road, and data packets are vehicles carrying small pieces of information.

2. The Internet Is a "Network of Networks"

The Internet is the global technical infrastructure that connects many independent networks. It is described as the backbone that makes the Web possible. It is not one giant computer and it is not owned by one organisation. Instead, universities, companies, governments, mobile operators, cloud providers, Internet Service Providers (ISPs), and many other networks interconnect using agreed communication standards.

2.1 Internet versus World Wide Web

The terms Internet and Web are often used as though they mean the same thing, but they are different:

Internet: The underlying worldwide network infrastructure that allows connected devices to exchange data.

World Wide Web (Web): A service that runs on the Internet. It consists of linked resources such as web pages, images, scripts, and data that are addressed by URLs and commonly transferred using HTTP/HTTPS.

Other services can also use the Internet, which is why the Internet is larger than the Web. For this bootcamp, our focus is the Web.

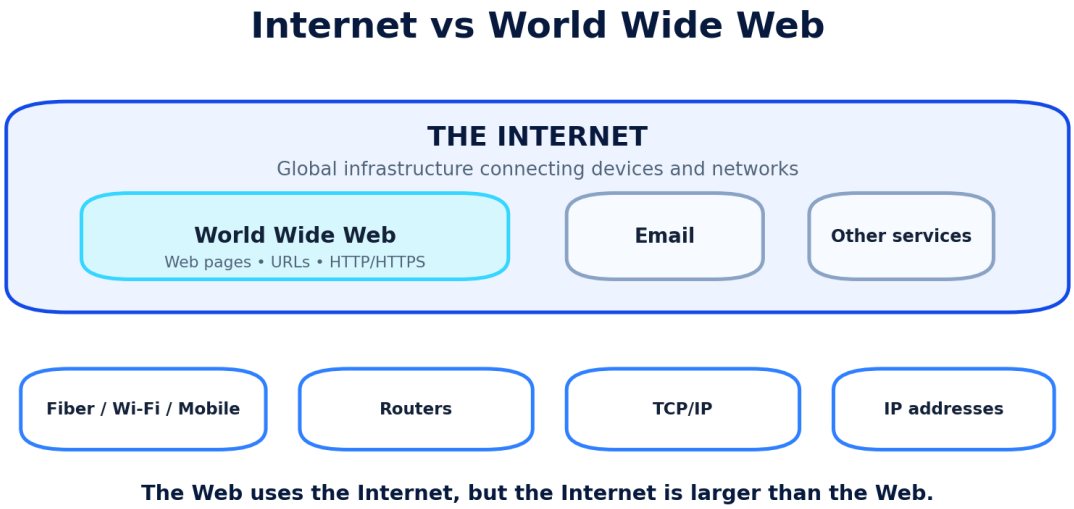


Figure 2. The Web is one major service that uses the Internet infrastructure.

3. The Main Players: Client, Browser, Server and Network Devices

3.1 Client

A client is a device or software that requests a service or resource from another computer. In normal web browsing, your laptop or phone is the client device, while Chrome, Firefox, Edge, Safari, or another browser acts as the web client software.

3.2 Browser

A web browser requests resources from web servers and then interprets them for the user. It reads HTML to understand the page structure, applies CSS to determine presentation and layout, and executes JavaScript to add behavior and interactivity. The browser is therefore where most front-end code runs.

3.3 Web server

A web server can refer to both the computer hosting a website and the server software that understands web requests. When the browser asks for a resource, the web server processes the request and returns a response. A static server may simply return stored HTML, CSS, JavaScript, or image files. A dynamic server may run application code, query a database, authenticate a user, and generate a response.

3.4 Router, switch, access point and ISP

Router: Connects networks and forwards packets toward their destination. Your home Wi-Fi router usually connects your local network to your ISP.

Switch: Connects multiple wired devices inside the same local network and forwards frames to the correct local device.

Wireless access point: Allows devices to join a local network using Wi-Fi.

ISP (Internet Service Provider): Provides your network with connectivity to the wider Internet. Mobile network operators can also provide Internet access.

4. IP Addresses, Domain Names and DNS

4.1 IP address

An IP address identifies a device or network interface so data can be routed to the correct destination. IPv4 addresses are often written as four decimal numbers separated by dots, while IPv6 uses a longer hexadecimal notation. At this stage, learners do not need to calculate addresses; the important idea is that computers need machine-readable addressing to deliver data.

4.2 Domain name

Humans prefer memorable names such as example.com rather than numerical IP addresses. A domain name is a human-friendly name associated with Internet resources.

4.3 DNS — Domain Name System

DNS works like a distributed directory. When you enter a domain name, the browser and operating system use DNS to find the IP address associated with the destination. Once the IP address is known, the client can send network traffic toward the correct server.

> **ANALOGY**
>
> If the domain name is a person's name in your contacts, DNS is the process of looking up that person's phone number. The name is easy for you to remember; the number is what the system needs to make the connection.

5. URLs: The Address of a Web Resource

A URL (Uniform Resource Locator) tells the browser which resource to access and includes information about how and where to access it. A URL can include a scheme/protocol, subdomain, domain, path, query string, and fragment.

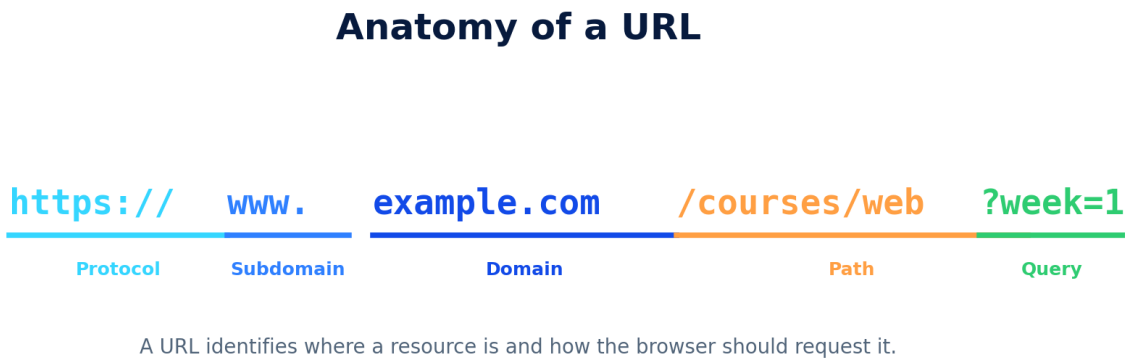


Figure 3. Simplified parts of a URL. Not every URL contains every optional part.

https:// Scheme / protocol: indicates HTTPS, the secure version of HTTP.

www. A subdomain. "www" is common but not compulsory.

example.com Domain name: the human-readable address of the host/site.

/courses/web Path: identifies a resource or route on the site.

?week=1 Query string: passes additional parameters to the server or application.

6. Packets, TCP/IP and Routers

When data moves across the Internet, it is not normally transmitted as one enormous indivisible block. Network communication breaks data into smaller units called packets. Each packet carries enough information for network systems to move it toward its destination. Routers examine addressing information and forward packets from one network to another.

At a high level, the TCP/IP family of protocols provides the rules that allow different computers and networks to communicate. For this bootcamp, remember two simplified responsibilities:

- IP helps address and route packets across networks.
- TCP is commonly used to provide reliable, ordered delivery for many Internet applications, including traditional HTTP connections.

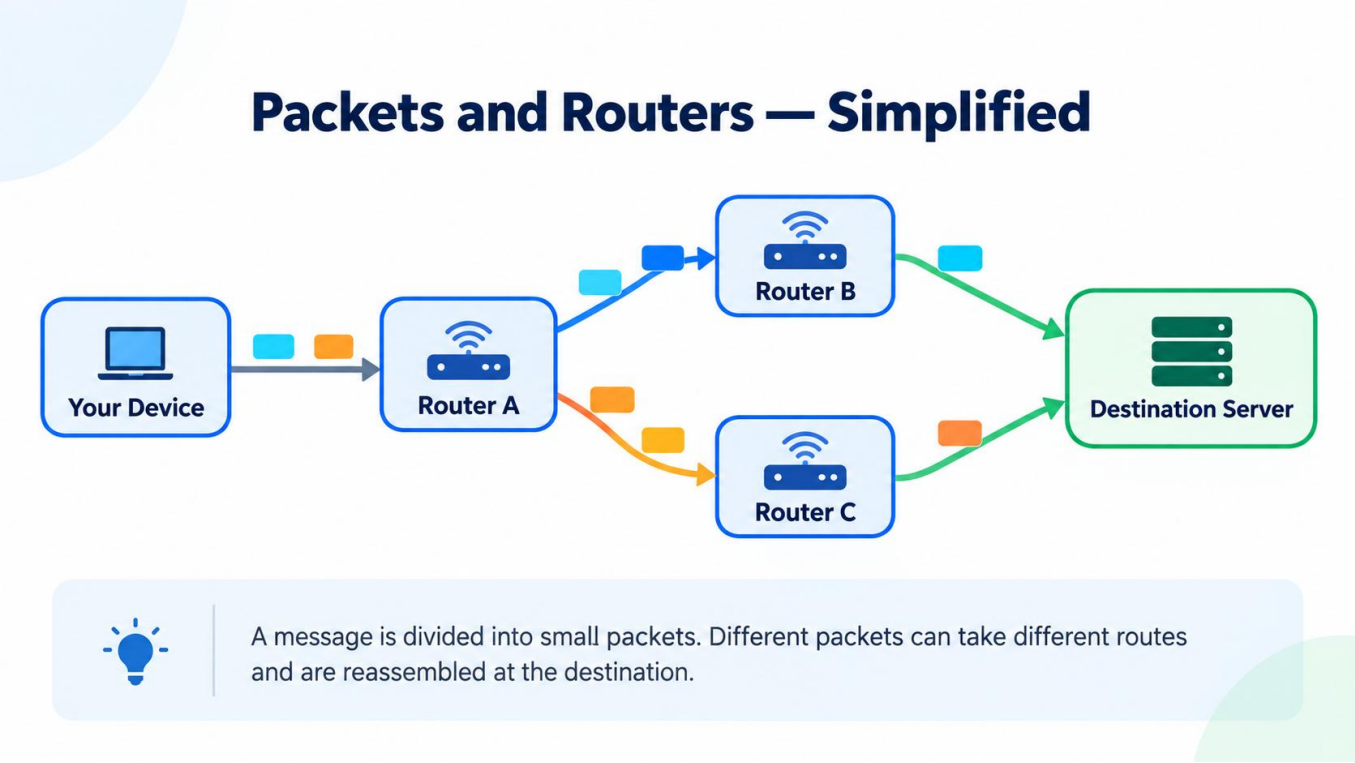


Figure 4. Packets can travel through intermediate routers before reaching the destination server.

DO NOT OVER-MEMORISE

You do not need to become a network engineer to build websites. The goal is to understand enough networking to make sense of URLs, servers, requests, hosting, deployment, APIs, errors, and browser–server communication.

7. HTTP and HTTPS: How the Browser and Server "Talk"

HTTP (Hypertext Transfer Protocol) is the application-layer protocol used for communication between web clients and servers. The browser sends an HTTP request; the server returns an HTTP response. We describe HTTP using the classic client–server model: a client opens communication, sends a request, and waits for a response.

7.1 Common HTTP methods

Method	Typical meaning	Simple example
GET	Retrieve/read a resource	Open a page or fetch a list of products
POST	Send data / create something	Submit a registration form
PUT/PATCH	Update a resource	Edit a profile
DELETE	Remove a resource	Delete a saved record

7.2 HTTP status codes

The response includes a status code. You will encounter these codes frequently as a web developer:

200 OK: The request succeeded.

301/302 Redirect: The requested resource is available at another location.

400 Bad Request: The server could not process the request as sent.

403 Forbidden: The server understood the request but refuses access.

404 Not Found: The requested resource could not be found.

500 Internal Server Error: The server encountered an error while processing the request.

7.3 HTTPS

HTTPS is HTTP protected by TLS encryption. It helps protect information while it travels between the browser and the server and helps the browser verify the server's identity through certificates. Modern production websites should use HTTPS, especially when users submit passwords, personal information, or payment details.

8. Step by Step: What Happens When You Open a Website?

Suppose a learner types https://example.com into a browser and presses Enter. A simplified sequence is:

1. Parse the URL — The browser identifies the HTTPS scheme, the domain name, and the requested path.

2. Resolve the domain — DNS is used to find the IP address of the server that hosts the site.

3. Establish network communication — The client communicates with the destination across the Internet. Routers forward packets between networks.

4. Secure the connection — For HTTPS, the client and server establish a protected TLS connection.

5. Send an HTTP request — The browser asks the server for the requested resource, often using GET.

6. Server processes the request — The server may simply return a file or may execute back-end code and query a database.

7. Server sends an HTTP response — The response includes a status code, headers, and usually a response body such as HTML or JSON.

8. Browser loads additional resources — HTML can reference CSS, JavaScript, images, fonts, and other files, causing more requests.

9. Browser renders the page — The browser interprets the received resources and displays the resulting interface to the user.

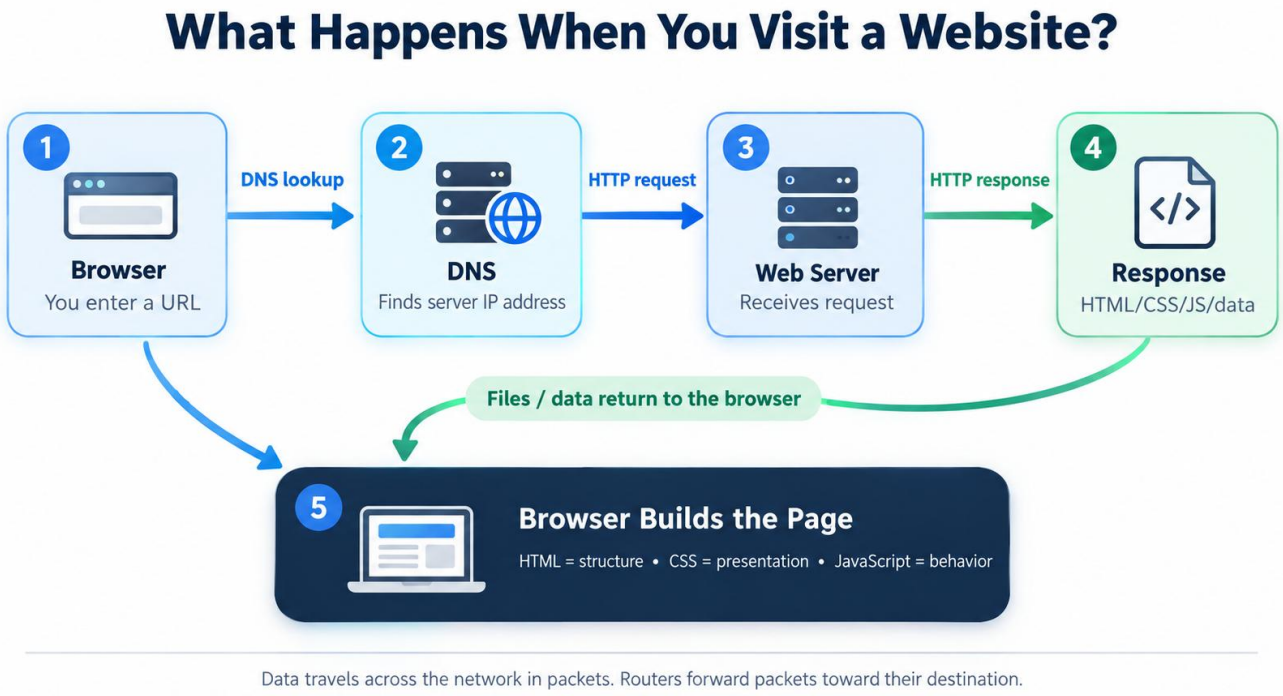


Figure 5. Simplified request–response path from the browser to the server and back.

9. Where Exactly Does Web Development Fit?

Now we can place the technologies in this bootcamp into the system. Web development is the creation of websites and web applications that users access through browsers. The work is commonly divided into front-end and back-end development.

9.1 Front-end development (client side)

Front-end code runs primarily in the user's browser and controls what the user sees and interacts with. In this cohort you will focus on:

HTML: Defines the meaning and structure of the page. MDN describes HTML as the most basic building block of the Web.

CSS: Controls presentation: layout, spacing, typography, colors, responsiveness, and visual styling.

JavaScript: Adds behavior and interactivity, such as reacting to clicks, validating forms, modifying the page, and communicating with APIs.

9.2 Back-end development (server side)

Back-end code runs on the server. It performs application logic that should not live solely in the browser. In Cohort 2, the planned technologies are PHP and MySQL. Typical back-end responsibilities include:

- Processing forms and requests.
- Authenticating users and managing sessions.
- Applying business rules.
- Reading and writing information in a database.
- Returning HTML, JSON, files, or other responses to the client.
- Protecting sensitive operations that should not be trusted to client-side code alone.

9.3 Database

A database stores structured application data. MySQL is a relational database management system. The browser normally does not connect directly to the production database; back-end application code communicates with the database and then returns only the appropriate data to the client.

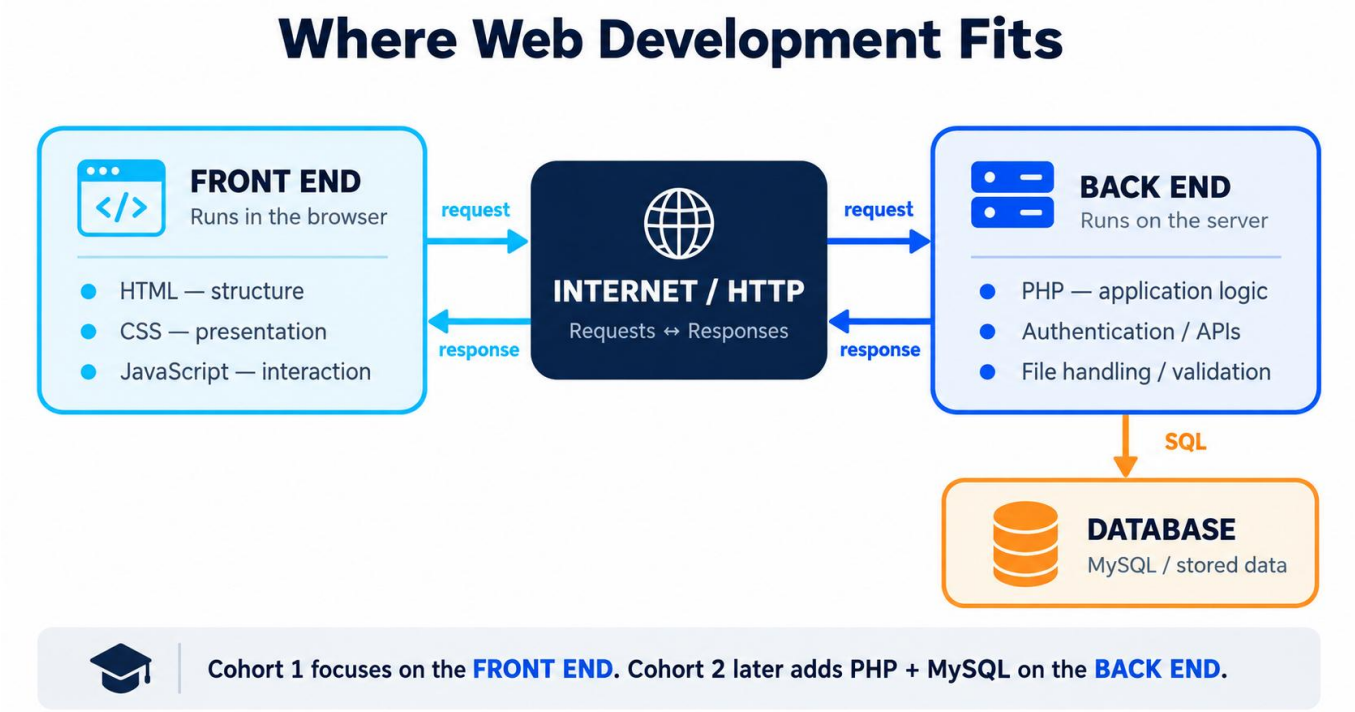


Figure 6. Web development sits above Internet networking: front end in the browser, back end on the server, and data in the database.

10. Why We Will Start Coding with Semantic HTML

The next lecture moves from infrastructure to the first front-end technology: HTML. MDN defines HTML as the basic building block of the Web and explains that it gives web content meaning and structure. This cohort will emphasise Semantic HTML from the beginning.

Semantic HTML means choosing elements according to the meaning and role of the content rather than using generic containers for everything. Examples include:

Semantic element	Purpose
<header>	Introductory content for a page or section.
<nav>	A section containing major navigation links.
<main>	The dominant main content of the document.
<section>	A thematic grouping of content, usually with a heading.
<article>	Self-contained content that could stand independently.
<aside>	Related or supplementary content.
<footer>	Footer information for a page or section.

THE BRIDGE TO LECTURE 2

Networks deliver the files. HTTP carries the request and response. The browser receives the response. HTML is the first language we will use to describe the structure and meaning of the page the browser receives.

11. Where This Lecture Sits in Your Bootcamp

Stage	Duration	Main focus	Technologies
Cohort 1	8 weeks	Front-end development	Semantic HTML, CSS, JavaScript
Project / transition break	2 weeks	Consolidate projects and prepare	Review, portfolio work, deployment

12. Review Questions

1. What is a computer network?
2. What is the difference between a LAN and the Internet?
3. Why is the World Wide Web not the same thing as the Internet?
4. What is the role of a browser in a web application?
5. What is the role of a web server?
6. Why do websites use domain names if computers use IP addresses?
7. What does DNS do?
8. What is a URL? Name at least three parts of a URL.
9. What is HTTP and what is the difference between a request and a response?
10. What does a 404 status code usually mean?
11. Why is HTTPS important?
12. What happens, at a high level, when a browser opens a website?
13. Where do HTML, CSS, and JavaScript normally run?
14. Where would PHP normally run?
15. Why should a browser usually not connect directly to a production database?
16. What does “Semantic HTML” mean, and why will we use it from the beginning?

13. Lecture Summary

- A network is a collection of connected devices that exchange data.
- The Internet is a global network of networks; the Web is a service that operates on top of the Internet.
- Browsers are clients; web servers respond to browser requests.
- DNS maps human-friendly domain names to IP addresses.
- Routers forward packets between networks.
- HTTP/HTTPS define how web clients and servers exchange requests and responses.
- HTML, CSS, and JavaScript form the core front-end technologies; PHP and MySQL will later support server-side logic and persistent data.
- This networking model explains what happens around the code you will start writing in the next lecture.

14. Key Terms Glossary

Term	Meaning
Browser	Software used to request, interpret, and display web content.
Client	A device or program that requests a resource or service.
Server	A computer or program that provides resources or services to clients.
Network	Connected devices capable of exchanging data.
Internet	The global system of interconnected networks.
Web	Linked resources and applications accessed over the Internet, commonly using HTTP/HTTPS.
IP address	A network address used to identify and route traffic to a destination.
DNS	Domain Name System; resolves domain names to IP addresses.
Domain name	Human-readable name associated with an Internet resource or host.
URL	Uniform Resource Locator; an address identifying a web resource.
HTTP	Protocol used for web requests and responses.
HTTPS	HTTP protected using TLS encryption.
Packet	A small unit of data transmitted across a network.
Router	Device/system that forwards packets between networks.
Front end	Browser-side part of a web application.
Back end	Server-side application logic and services.
Database	System used to store and retrieve application data.
Semantic HTML	HTML that uses elements according to the meaning/role of the content.